💡 Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime. ×

RESULTS

# Results

01 · FACTORIAL ANOVA

main effects + interaction of 2+ factors

**Table 1.** Cell descriptives

| Factor A × B | N | Mean | Std. Dev. |
|---|---|---|---|
| Blended × Female | 40 | 66.63 | 6.73 |
| Flipped × Female | 40 | 70.88 | 6.55 |
| Lecture × Female | 40 | 59.58 | 7.45 |
| Blended × Male | 40 | 64.36 | 6.86 |
| Flipped × Male | 40 | 70.62 | 6.69 |
| Lecture × Male | 40 | 61.34 | 7.67 |

**Table 2.** ANOVA (main effects + interaction)

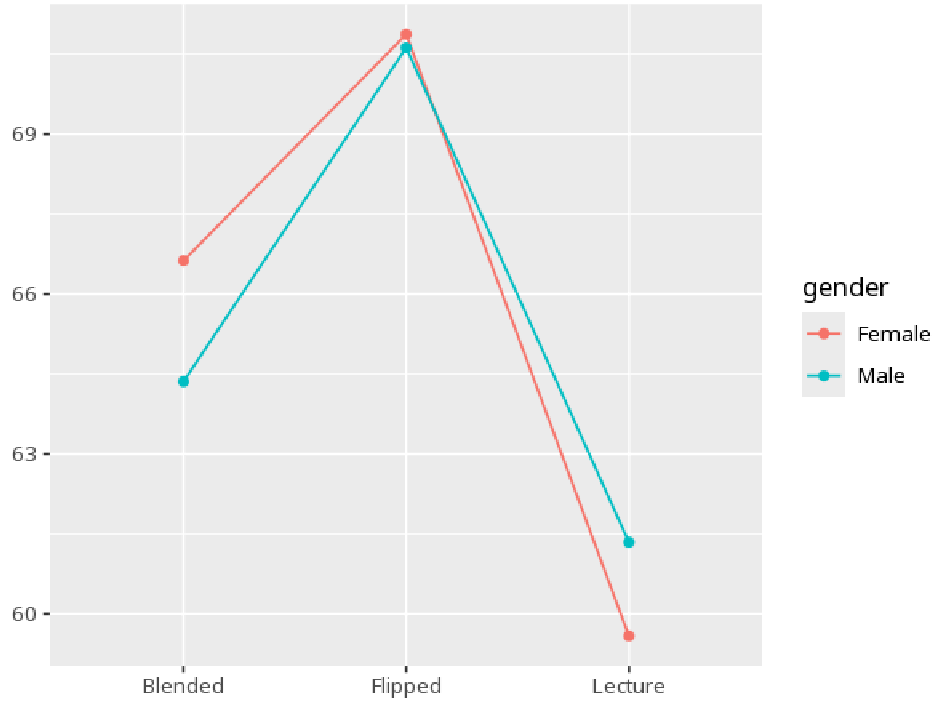
| Source | SS | df | MS | F | p | partial $\eta^2$ [95% CI] |
|---|---|---|---|---|---|---|
| teaching_method | 4234.40 | 2 | 2117.20 | 43.14 | <.001 | 0.27 [0.19, 1.00] |
| gender | 3.85 | 1 | 3.85 | 0.08 | .780 | 0.00 [0.00, 1.00] |
| teaching_method × gender | 162.25 | 2 | 81.12 | 1.65 | .194 | 0.01 [0.00, 1.00] |

**Table 3.** Simple effects / post-hoc

| Contrast | $M_{diff}$ | SE | $p_{adj}$ | 95% CI |
|---|---|---|---|---|
| Blended - Flipped | −5.26 | 1.11 | <.001 | [−7.87, −2.64] |
| Blended - Lecture | 5.03 | 1.11 | <.001 | [2.42, 7.64] |
| Flipped - Lecture | 10.29 | 1.11 | <.001 | [7.68, 12.90] |

assumption checks: Levene's & normality of residuals; post-hoc / simple-effects table when an effect is significant. (Levene F=0.52, p=.761 · Shapiro W=0.99, p=.022) - equal variances look reasonable - residual normality looks doubtful; interpret the F-tests with extra caution

**Figure.** Interaction




### Understanding the numbers

**Source.** Names which effect this ANOVA row tests - a main effect for one factor, or the A×B interaction between them. Here, the headline effect discussed above is teaching_method × gender.

**SS.** The portion of total variability in the outcome attributable to that row's effect. Here, teaching_method × gender has SS = 162.25.

**df.** The degrees of freedom for that effect - roughly, the number of independent levels being compared. Here, teaching_method × gender has df = 2.

**MS.** The effect's SS divided by its df - the average variability per degree of freedom. Here, teaching_method × gender has MS = 81.12.

**F.** The ratio of an effect's mean square to the residual (error) mean square - larger means the effect explains more variance than chance predicts. Here, F for teaching_method × gender = 1.65.

**p.** The probability of an effect this large (or larger) if that factor/interaction truly had no influence. Here, p = .194 for teaching_method × gender - at or above the $\alpha$ = 0.05 threshold.

**partial $\eta^2$.** The share of variance that effect explains after removing variance due to the other effects (~.01 small, .06 medium, .14 large). Here, partial $\eta^2$ for teaching_method × gender = .01 [.00, 1.00].

**Cell (Factor A × B).** Each row of the cell-descriptives table names one specific combination of factor levels. Here, the cell descriptives table above breaks the data into 6 such factor-level combinations.

**N.** The number of observations in that cell. Here, each of the 6 cells above has its own N (see the cell descriptives table for each count).

**Mean.** The arithmetic average of the outcome within that cell. Here, each cell's mean is listed separately in the cell descriptives table above.

**Std. Dev..** How spread out the outcome scores are within that cell. Here, each cell's SD is listed separately in the cell descriptives table above.

**M diff.** The difference between two cell/level means being compared in the post-hoc / simple-effects table. Here, each row of the simple-effects table above reports its own mean difference for a specific pair.

**SE.** The standard error of that mean difference. Here, each contrast's SE is listed alongside its mean difference in the simple-effects table above.

**p (adj).** The pairwise p-value, Tukey-adjusted for the number of comparisons made. Here, each pairwise contrast above carries its own Tukey-adjusted p.

**95% CI.** The range likely to contain the true mean difference for that pair. Here, each contrast's 95% CI is shown in the simple-effects table above.

### How to read this test

Each factor's F/p is its main effect; the A×B row tests whether the effect of one factor depends on the other. A significant interaction usually takes priority - read it from the interaction plot before the main effects. A significant effect for any factor with 3+ levels tells you the levels differ somewhere, not which ones - use the post-hoc / simple-effects (emmeans) table to find the specific pairs. Your significance threshold (α) is 0.05.

**APA template:** A two-way ANOVA gave A×B interaction $F(2,234)=1.65$, $p = .194$, partial $\eta^2=.01$ [.00, 1.00].

**Statistical basis:** Factorial ANOVA (main effects + interaction) (Fisher, R. A. (1925). Statistical Methods for Research Workers. Oliver & Boyd.); Estimated marginal means for interaction / simple-effects follow-up (Lenth RV (2024). emmeans: Estimated Marginal Means, aka Least-Squares Means. R package.); $\eta^2$ effect-size benchmarks (Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Routledge.). Full references in the exported CITATIONS.txt.